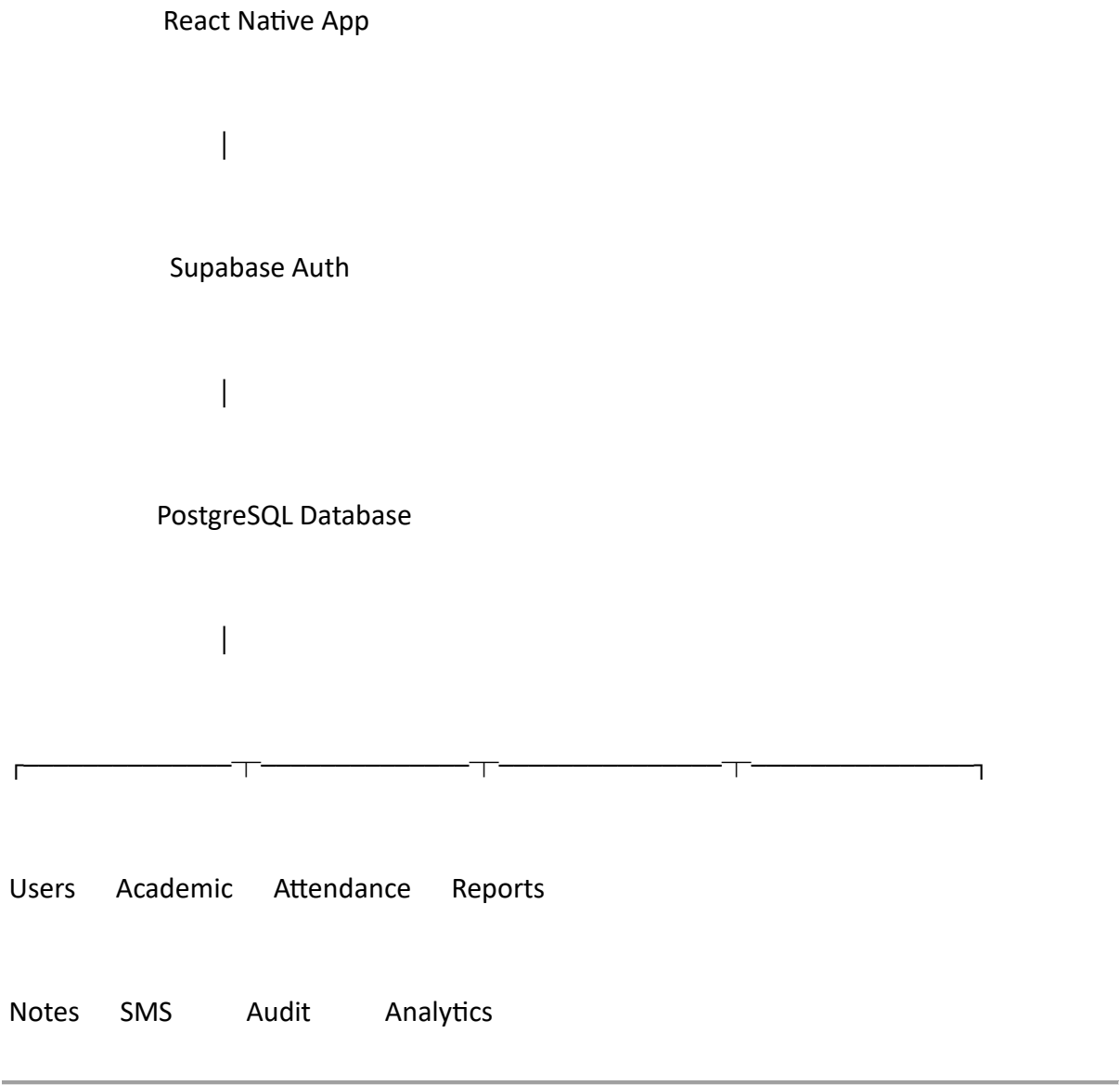


VDC Teacher Application Backend Schema (Version 1)

Architecture



Database Modules

Authentication

Academic Management

Attendance

Assessment

Reports

Notes

SMS

Settings

Audit Logs

Entity Relationship Overview

Teacher

|

|— TeacherClass

| |

| ▼

| Class

| |

| |— Student

| |

| |----- Attendance

| |

| |----- Test

| |

| |----- Assignment

|

Notes

SMS Logs

Reports

Authentication Module

users

This table represents every authenticated user.

id UUID PK

auth_user_id UUID

employee_code

first_name

last_name

email

phone

role_id

department_id

status

profile_photo

created_at

updated_at

deleted_at

roles

id

role_name

description

Examples

Teacher

Admin

Principal

Department

departments

id

department_name

department_code

created_at

Semester

semesters

id

semester_number

academic_year

status

Section

sections

id

semester_id

section_name

strength

Subject

subjects

id

subject_code

subject_name

credits

department_id

semester_id

Teacher Assignment

A teacher can handle multiple classes.

teacher_classes

id

teacher_id

subject_id

semester_id

section_id

academic_year

created_at

Student

students

id

usn

roll_number

first_name

last_name

gender

date_of_birth

phone

parent_phone

email

semester_id

section_id

department_id

status

created_at

updated_at

Attendance Module

attendance_sessions

One attendance record per class.

id

teacher_class_id

attendance_date

period

remarks

created_by

created_at

attendance_details

id

attendance_session_id

student_id

status

marked_at

Status

Present

Absent

Late

OD

Leave

SMS Module

sms_logs

id

attendance_session_id

student_id

phone_number

message

provider

status

response

sent_at

Status

Pending

Success

Failed

Test Module

tests

id

teacher_class_id

title

max_marks

test_date

description

created_at

test_marks

id

test_id

student_id

marks_obtained

remarks

entered_at

Assignment Module

assignments

id

teacher_class_id

title

max_marks

submission_date

description

created_at

assignment_marks

id

assignment_id

student_id

marks_obtained

remarks

entered_at

Notes Module

notes

id

teacher_id

subject_id

title

description

file_name

file_type

file_url

file_size

uploaded_at

Storage

Supabase Storage

Only URL stored here.

Reports

Instead of tables,

use SQL Views.

vw_monthly_attendance

student_id

month

present

absent

percentage

vw_quarterly_attendance

student_id

quarter

present

absent

percentage

vw_test_summary

student

highest

lowest

average

vw_assignment_summary

student

highest

lowest

average

Notification Module

Future

notifications

id

user_id

title

message

type

status

created_at

Audit Module

Every table

created_by

created_at

updated_by

updated_at

deleted_at

Soft delete only.

Storage Buckets

Supabase

teacher-notes

profile-images

assignment-files

report-exports

Indexes

Students

USN

Roll Number

Semester

Section

Attendance

attendance_date

teacher_class

student

Tests

teacher_class

test_date

Assignments

teacher_class

Relationships

Role



Users



Teacher



Teacher Class



Class



Students



Attendance



Tests



Assignments



Reports

Row Level Security (Supabase)

Teachers should only access the classes assigned to them.

Example policy:

Teacher can SELECT teacher_classes

WHERE teacher_id = auth.uid()

Attendance, marks, notes, and reports should all be filtered through the teacher's assigned classes. Administrators can be granted broader policies later without changing the schema.

API Resources (Logical)

/auth

/classes

/students

/attendance

/tests

/assignments

/reports

/notes

/sms

/profile

Recommended Folder Structure (Backend Logic)

database/

├─ migrations/

├─ seed/

├─ functions/

├─ policies/

├─ triggers/

├─ views/

└─ backups/

Future Expansion

This schema is intentionally designed so you can add new features without redesigning the database:

- Parent Portal
- Student Portal
- Admin Portal
- Timetable Management

- Fee Management
- Examination Management
- Push Notifications
- AI Analytics
- OCR Attendance Import
- Offline Synchronization
- Multi-campus / Multi-college support

One architectural improvement I'd recommend

Since this is a **college application**, I would introduce two additional core entities that simplify future growth:

- **Academic Year** (e.g., 2026–2027)
- **Course/Program** (e.g., BCA, BCom, BSc)

Then your hierarchy becomes:

Academic Year

|

Course / Program

|

Semester

|

Section

|

Subject

|

Teacher Class

|

Students

|

Attendance / Tests / Assignments

This structure supports multiple academic years, different programs, and future expansion to additional departments or institutions without requiring major schema changes. It is the approach commonly used in scalable Student Information Systems (SIS) and ERP platforms.